



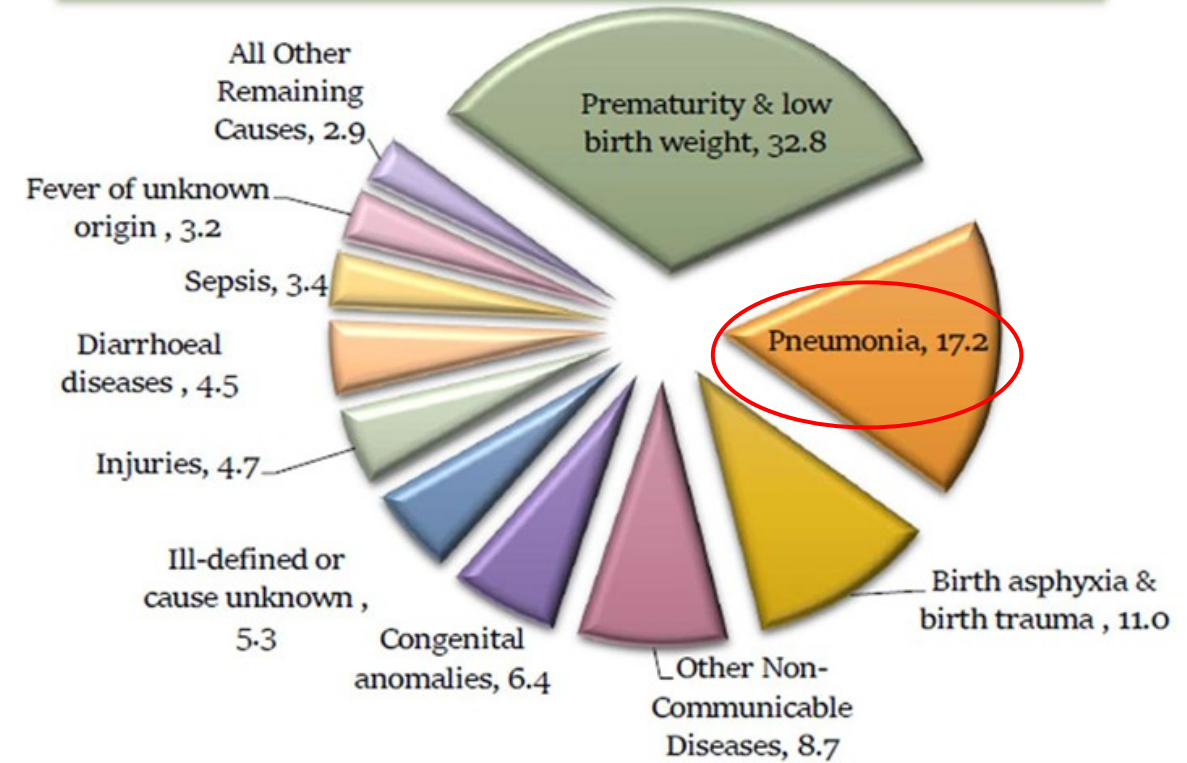
TOPIC 6

Identification & Management of Childhood Pneumonia

**Presenter: Dr. Lingaraj
Pradhan**



Chart 33 - Distribution of % proportion of deaths for age group 0-4 year in India, 2022-2024 (in %)



Childhood Pneumonia

Pneumonia is a **leading cause of U5 deaths** in the State

Childhood Pneumonia: Why It Matters

Leading Cause of Under-5 Deaths

Pneumonia remains a leading cause of death in children under 5 years of age in the State, making early identification and correct classification a frontline priority for every health worker.

Identification is divided into two age groups:



0-2 Months

Classified as Possible Serious Bacterial Infection (PSBI)



2 Months - 5 Years

Classified as No Pneumonia, Pneumonia, or Severe Pneumonia

PSBI Danger Signs in Young Infants

Possible Serious Bacterial Infection (PSBI) —
classify if **ANY ONE** of these signs is present



Not able
to feed



Convulsions



Fast breathing —
60 breaths/min
or more



Severe chest
indrawing



Axillary temperature
37.5°C or above
(feels hot)



Axillary temperature
below 35.5°C
(feels cold)



Movement only
when stimulated,
or no movement
at all



Grunting



Bilateral
nasal flaring



If **ANY ONE** sign is present,
classify as PSBI and seek urgent medical care.



Pre-Referral Care & Antibiotic Therapy for PSBI

PRE-REFERRAL STEPS

- Warm the infant by skin-to-skin contact with mother/caregiver
- Treat to prevent low blood sugar
- If able to breastfeed, ask mother to breastfeed the child
- If not able to breastfeed, give 20–50 ml (10 ml/kg) expressed breast milk by OG tube, or 2 ml/kg of 10% dextrose
- Give first pre-referral dose of Inj. Gentamicin + Syp. Amoxicillin, maintain warmth, and refer



If weight is below 1.5 kg, refer to a higher facility directly — do not delay.

ANTIBIOTIC DOSING BY WEIGHT

Weight	Gentamicin (80mg/2ml vial)	Amoxicillin Syrup (125mg/5ml)	Amoxicillin in Tablet (250mg)
< 1.5 kg	Refer to higher facility directly		
1.5 – 2.0 kg	0.2 ml	2 ml	¼ tab
2.0 – 3.0 kg	0.3 ml	2.5 ml	½ tab
3.0 – 4.0 kg	0.4 ml	3 ml	½ tab
4.0 – 5.0 kg	0.5 ml	4 ml	½ tab
Gentamicin: 5 mg/kg/dose, once a day. Amoxicillin: 25 mg/kg/dose, twice a day.			

Classifying Pneumonia in Children

Any child presenting with fever, cough, or cold must be assessed for the following classification:

SEVERE PNEUMONIA

- General danger signs:
can't feed/drink, vomits everything, lethargy, reduced consciousness, convulsions
- Central cyanosis or $\text{SpO}_2 < 90\%$
- Severe respiratory distress ($\text{RR} > 70/\text{min}$)
- chest indrawing
- Head nodding, stridor,

PNEUMONIA

- Fast breathing:
 - 2-11 months: $\text{RR} \geq 50/\text{min}$
 - 12-59 months: $\text{RR} \geq 40/\text{min}$

NO PNEUMONIA

No signs of Pneumonia or Severe Pneumonia are present.

Remember: classify using the highest level of severity for which the child has even one sign present.

PNEUMONIA • NO PNEUMONIA

Home Care for Cough and Cold (No Pneumonia)

ADVICE TO THE MOTHER

- Keep the child properly clothed and warm
- Continue feeding during illness; breastfeeding should be continued
- Give home-available fluids as much as the child will take — helps relieve cough
- For children above 1 year, give a home-made soothing remedy (honey, ginger, herbal concoctions). No remedy for an exclusively breastfed child

Encourage the mother to keep feeding and offering fluids throughout the illness.

- Keep the nose clean using nasal drops

DANGER SIGNS -REPORT TO ASHA / ANM IMMEDIATELY



- Child becomes sicker
- Not able to drink or breastfeed
- Fast or difficult breathing

Acting early for these signs can prevent the illness from worsening.

Treatment of Pneumonia with Oral Amoxicillin

TREATMENT PROTOCOL

- Give oral Amoxicillin 25 mg/kg/dose, twice a day, for 5 days
- Give first dose at clinic; teach mother to give remaining doses at home
- Advise to continue feeding
- Advise to return after 2 days, or earlier if child becomes sicker or is unable to drink/breastfeed
- Avoid unnecessary medication: cough syrups, medicated nose drops, steam



Complete the full 5-day course even if the child improves earlier.

AMOXICILLIN DOSING — TWICE DAILY FOR 5 DAYS

Age / Weight	Syrup (125mg/5ml)	Dispersible Tablet (250mg)
2-4 months (4 to <6 kg)	5 ml	½ tab
4-12 months (6 to <10 kg)	10 ml	1 tab
12 mo – 3 yrs (10 to <14 kg)	15 ml	1½ tab
3 – 5 years (14 to <20 kg)	—	2 tab

Note: Use weight to select the dose whenever the child's weight is known — it is more accurate than age alone, especially for children with malnutrition.

SEVERE PNEUMONIA

Severe Pneumonia: Pre-Referral & Wheeze Management

PRE-REFERRAL TREATMENT

- Hospitalize the child
- Give oxygen if saturation is below 90% or Respiratory Rate $\geq$ 70/Minute
- Manage the airway

IF REFERRAL REFUSED OR NOT FEASIBLE

- Give recommended pre-referral

Manage illness using Oral Amoxicillin twice a day AND Injection Gentamicin once a day for 7 days, by CHO/ANM.

IF THE CHILD HAS WHEEZING

- Give 3 doses of nebulized Salbutamol for 20 minutes; or 4 puffs of Salbutamol MDI with spacer (gap of 5 min between puffs), repeated every 20 minutes
- Reassess the child and reclassify the problem
- No antibiotics needed if significant improvement with bronchodilator and no clinical signs of consolidation
- If tachypnoea, distress, toxic look, or consolidation persists — continue antibiotics along with bronchodilators

Management of Severe Pneumonia: Injectable Antibiotics

FIRST-LINE INJECTABLE ANTIBIOTICS

- Ampicillin 50 mg/kg IM or IV every 6 hours
- Gentamicin 7.5 mg/kg IM or IV once a day

SPECIAL SITUATIONS

- Staphylococcal infection suspected (skin pustules/boils): give Cloxacillin OR Amoxicillin + Clavulanic acid
- Septic shock: give Ceftriaxone with Vancomycin
- No improvement within 48 hours: switch to Gentamicin 7.5 mg/kg IV once daily + Ceftriaxone 100 mg/kg IV in two divided doses, OR Cloxacillin 50 mg/kg IV every 6 hours

SHIFTING TO ORAL & TOTAL DURATION

Shift to oral drugs as soon as the child can take orally: Inj. Ampicillin → oral Amoxicillin; Inj. Gentamicin IV → IM, to complete the antibiotic course.

Clinical Scenario	Total Antibiotic Duration
Clinical response within 48 hours	7 days
Clinical response after 48 hours	10 days
Empyema	4-6 weeks

Oxygen Therapy & Nebulization

OXYGEN THERAPY



- Give oxygen if saturation $< 90\%$ (or $< 94\%$ if other danger signs like shock are present)
- Use nasal prongs at 0.5–1 litre/min — preferred for young infants. If unavailable, use nasal or nasopharyngeal catheter
- Use pulse oximetry to guide therapy; keep saturation above 90%
- If pulse oximeter unavailable, continue oxygen until clinical signs of hypoxia resolve (e.g. unable to

NEBULIZATION WITH SALBUTAMOL

- Add salbutamol to nebulizer cup with saline to make volume 3 ml — dose 2.5 mg (0.5 ml of the 5mg/ml solution)
- Give every 1–4 hours initially, reducing to 6–8 hourly as the child improves

Oral/Syrup Salbutamol: < 1 yr — 1 mg/dose; 1–4 yrs — 2 mg/dose, both every 6–8 hourly.

In severe pneumonia, do NOT delay antibiotic administration — even while giving bronchodilators.

Supportive Care, Monitoring, Discharge & Follow-up

SUPPORTIVE CARE

- Clear thick nasal/throat secretions by gentle suction
- Give paracetamol for fever $\geq 38.5^{\circ}\text{C}$
- Provide maintenance IV fluids only if unable to feed orally; stop gradually once tolerating
- Give rapid-acting bronchodilator if wheeze is present
- Encourage feeding as soon as possible — no role for cough syrup (may be harmful)

● Ensure vaccination and nutrition advice

Arrange follow-up 2 weeks after discharge — check nutrition, vaccination, and address risk factors (malnutrition, indoor air pollution, parental smoking).

MONITORING FREQUENCY

Nurse: at least every 3 hours **Doctor:** at least twice a day

Improvement signs within 48 hrs: slower breathing, less chest indrawing, less fever, better feeding, improved SpO₂.

DISCHARGE CRITERIA

- Respiratory distress resolved; SpO₂ $\geq 90\%$ on room air
- Feeding well
- Able to take oral medication, OR has completed a course of parenteral antibiotics
- Family counseled on when to return